American Computer Science League

2023-2024 ● Contest 4: Short Problems Solutions ● Classroom Division

1. Graph Theory

The adjacency matrix raised to the 2nd power for paths of length 2 is as follows:

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} 0 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{bmatrix}$$

There are 15 entries of 0.

C. 15

2. Graph Theory

By inspection, there are 3 cycles: BDB, ACEA, ABDA.

B. 3

3. Digital Electronics

The Boolean expression for the digital circuit is: $\overline{\overline{\overline{A} + AB} + \overline{BC}}$

$= \overline{\overline{\overline{A} + AB}}\,\overline{\overline{BC}}$

$= (\overline{A} + AB)(BC)$

$= \overline{A}BC + ABC$

$= ABC$

D. ABC

4. Digital Electronics

The Boolean expression for the digital circuit is: $\overline{AB\,\overline{\overline{BC} + C}}$

$= \overline{AB} + \overline{\overline{BC} + C}$

$= \overline{AB} + \overline{BC} + C$

$= \overline{A} + \overline{B} + \overline{B} + \overline{C} + C$

$= 1$

Therefore all ordered triples make it TRUE which is 8 of them.

E. 8

5. Assembly Language

This program finds the sum of the proper factors of the number NUM which are 1, 2, 4, 7, 14. 28 is a perfect number so the sum is the same as the number.

C. 28

6. Graph Theory

The adjacency matrix for the graph is:

$$\begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 \end{bmatrix}^2 = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 1 \\ 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 2 & 0 & 1 & 1 \end{bmatrix}$$

Summing the entries in the squared matrix gives the number of paths of length 2 which is 18.

E. 18

7. Graph Theory

By inspection, there are 19 total cycles, but only 11 originate at vertex A. Remember that ACEA and AECA are the same cycle so the extra 8 of them don't include vertex A at all. They are:
ABA, AEA, ABEA, ACBA, ACEA, AEBA, ABCEA, ACBEA, ACEBA, ABDCEA, AEDCBA

D. 11

8. Digital Electronics

This circuit translates to the Boolean expression: $\overline{AB} \oplus B\overline{C}$. The truth table for this is as follows:

A	B	C	$\overline{AB}$	$B\overline{C}$	$\overline{AB} \oplus B\overline{C}$
0	0	0	1	0	1
0	0	1	1	0	1
0	1	0	1	1	0
0	1	1	1	0	1
1	0	0	1	0	1
1	0	1	1	0	1
1	1	0	0	1	1
1	1	1	0	0	0

The ordered triples that make this TRUE are: 000, 001, 011, 100, 101, 110

D. 6

9. Digital Electronics

This circuit translates to the Boolean expression: $\overline{AB + B + \overline{B + C}}$.

$\overline{AB + B + \overline{B + C}} = \overline{AB} * \overline{B} * (B + C) = (\overline{A} + \overline{B}) * \overline{B} * (B + C)$
$\overline{A} * \overline{B} + \overline{B} * \overline{B} * (B + C) = \overline{B} * (B + C) = \overline{B} * B + \overline{B}C = \overline{B}C$

A. $\overline{B}C$

10. Assembly Language

This finds which triangular number 78 is. The previous ones are: 1, 3, 6, 10, 15, 21, 28, 36, 45, 55, 66, 78. It's the 12th number in the sequence.

B. 12